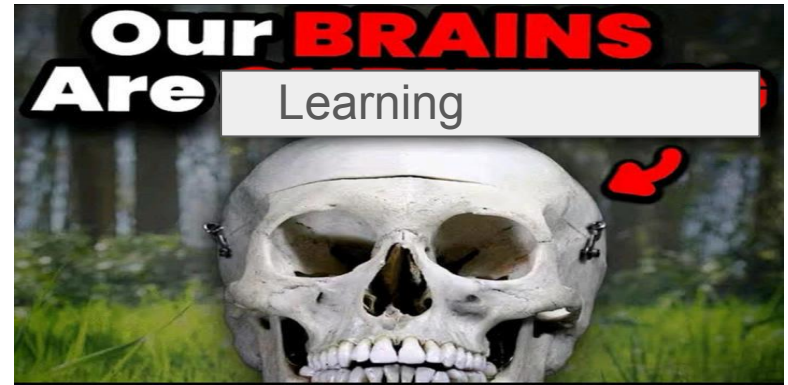




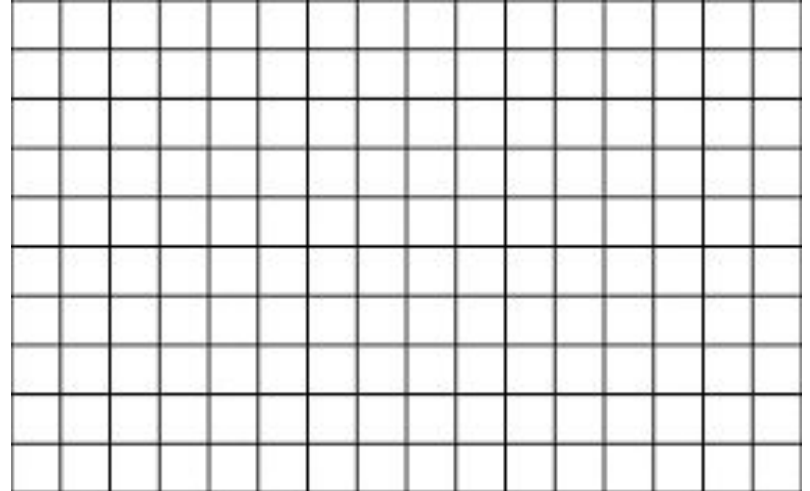
Conway's Game of life

Click to add subtitle



Rules

1. Any live cell with fewer than two live neighbours dies, as if by underpopulation.
2. Any live cell with two or three live neighbours lives on to the next generation.
3. Any live cell with more than three live neighbours dies, as if by overpopulation.
4. Any dead cell with exactly three live neighbours becomes a live cell, as if by reproduction.



Gaming time

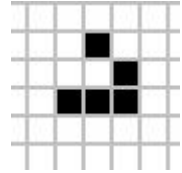
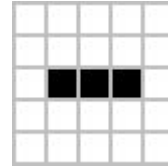
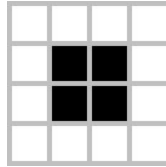
<https://playgameoflife.com/>

3 main types of structures

Still lifes

<https://playgameoflife.com/>

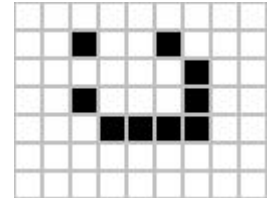
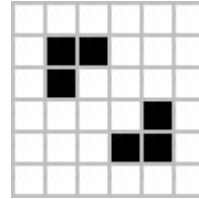
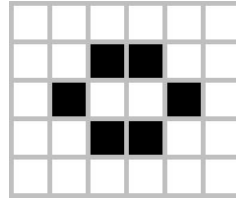
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Blinkers

<https://playgameoflife.com/lexicon/blinker>

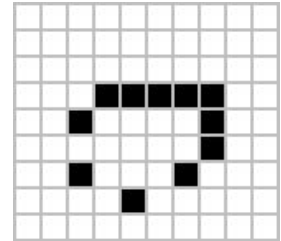
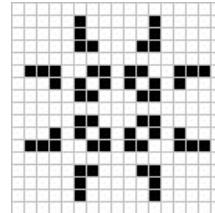
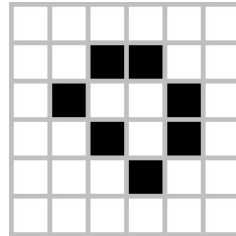
https://playgameoflife.com/lexicon/quasar_%281%29



Rocket ship

<https://playgameoflife.com/>

<https://playgameoflife.com/lexicon/295P5H1V1>



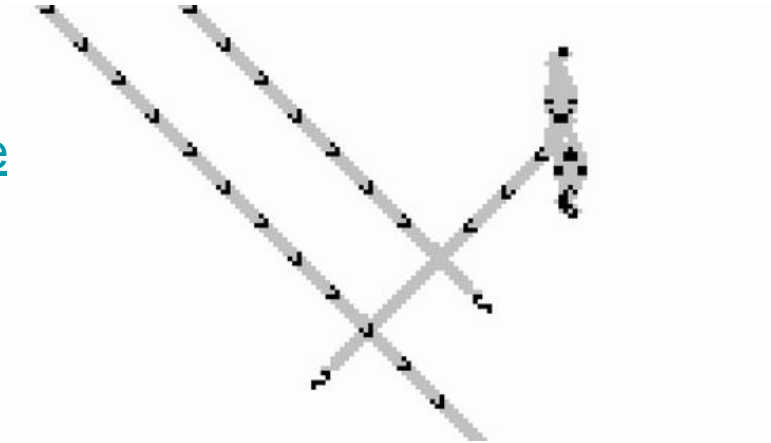
Other cool structure

Glider gun

https://playgameoflife.com/lexicon/Gosper_glider_gun

Glider Collision

<https://playgameoflife>



Turing completeness

1. Conditional logic

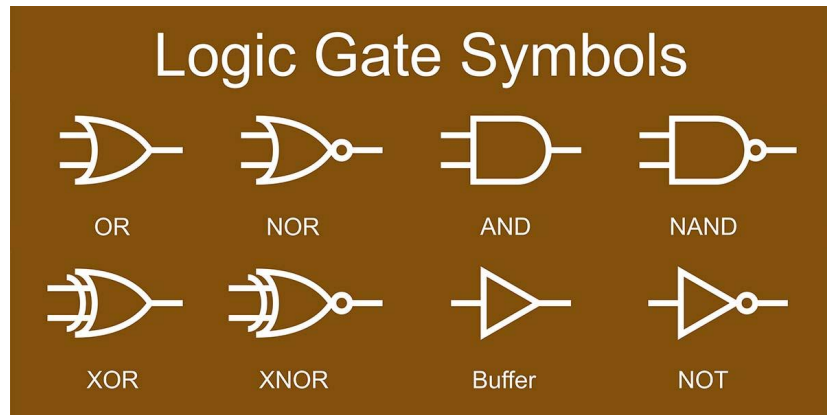
The ability to make decisions like:

- “If *X*, then do *Y*; otherwise do *Z*.”
(Examples: `if` statements, conditional jumps, pattern matches)

2. Infinite or unbounded memory

Not literally infinite—just **no fixed upper limit** determined by the language.

(Examples: ability to grow a list, allocate more memory, write to an unbounded tape)

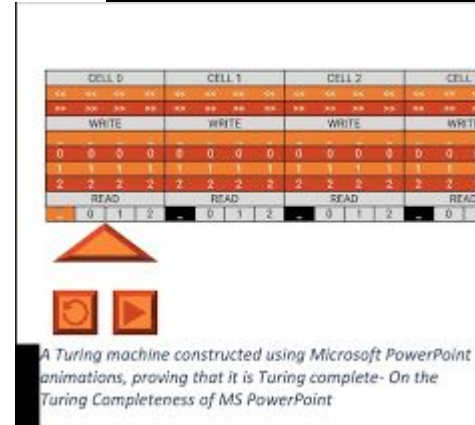
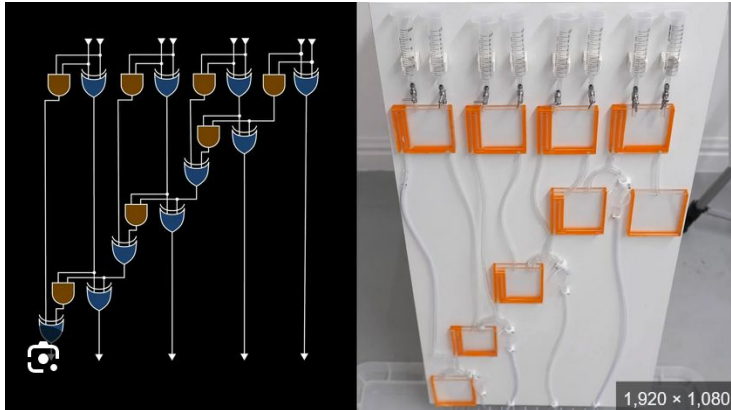
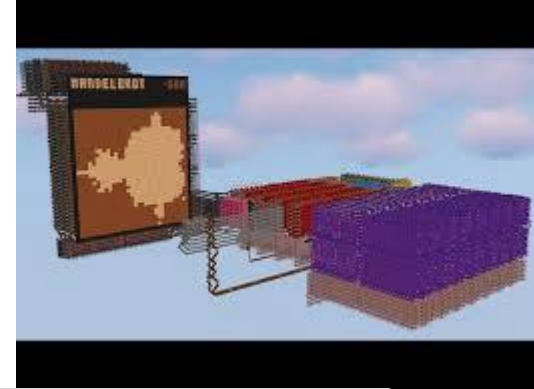
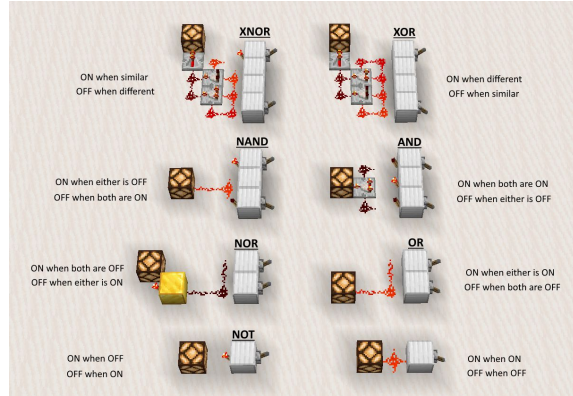


Some surprising things that are turing complete

Minecraft redstone

Water

Powerpoint



Conways game of life is Turing complete

<https://www.youtube.com/watch?v=xP5-ileKXE8>